UNIT 8: ECOLOGICAL CONCEPTS
8.7 – CONSERVATION OF BIODIVERSITY

Biology 9 Mr. Queenan

Ch. 6.3 (pg. 186-189) & Ch. 7.1-7.4 (pg. 202-225)

Conservation of Biodiversity Outline

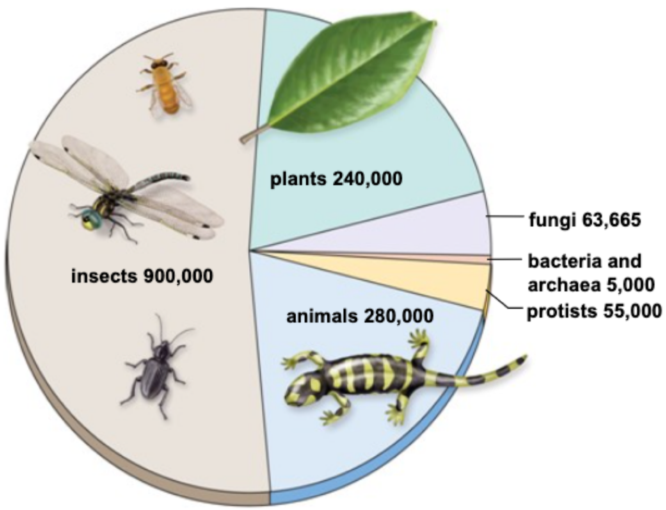
- Conservation Biology and Biodiversity
- Value of Biodiversity
- Causes of Extinction
- Conservation Techniques

Conservation Biology and Biodiversity

- **Conservation Biology** considers all aspects of biodiversity
  - General goal is conserving natural resources for this and future generations
  - Primary goal is the management of **biodiversity**
    - The variety of life on Earth
- For conservation biology to be effective, scientists must evaluate larger connections within the biosphere
  - High level of biodiversity is desirable
  - Causes of present-day extinction, how to prevent future extinctions from occurring, and consequences of reduced biodiversity
  - **Bioinformatics** is utilized to protect biodiversity
    - Collecting of, analyzing, and making readily available biological information, using modern computer technology

Conservation Biology and Biodiversity

- Biodiversity
  - At its simplest level, biodiversity refers to the variety of species on Earth
    - Estimated that between 10 and 50 million species currently exist
    - **Endangered Species**
      - One that is in peril of immediate extinction throughout all or most of its range
    - **Threatened Species**
      - Organisms that are likely to become endangered in the near future
  - Ecologists describe biodiversity as a combination of three levels of biological organization:
    - Genetic diversity
    - Community diversity
    - Landscape diversity

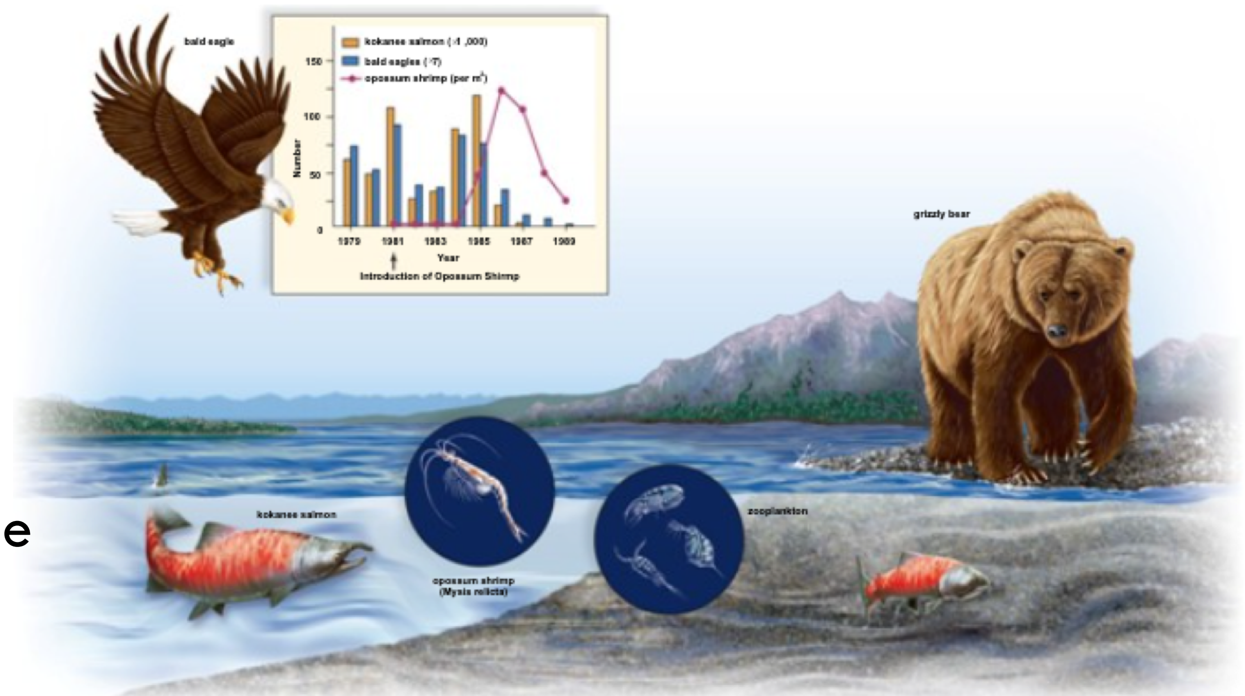


Conservation Biology and Biodiversity

- Biodiversity
  - **Genetic diversity** refers to variations among the members of a population
    - Populations with high genetic diversity are more likely to have some individuals that can survive a change in the structure of their ecosystem
    - If a species' population is small and isolated, it is more likely to become extinct due to a limited genetic diversity.
  - **Ecosystem diversity** is dependent on interactions of species in a particular area
    - A diverse community composition will increase the levels of biodiversity in the biosphere
    - An effective approach to conservation is to conserve species that play a key role within the ecosystem
      - Saving an entire community can save many species
  - **Landscape diversity** involves a group of interacting ecosystems within one landscape
    - **Landscape** – Ex: mountains, rivers, grasslands
    - Fragmentation of the landscape reduces reproductive capacity and food availability (building a road across a patch of land)

Conservation Biology and Biodiversity

- Ecosystem Diversity
  - Opossum shrimp introduced as a food source for salmon.
    - Ended up competing with salmon for zooplankton
    - Resulted in decline of salmon populations , as well as organisms that fed on salmon



Conservation Biology and Biodiversity

- Distribution of Biodiversity
  - Biodiversity is not evenly distributed throughout the biosphere
  - Biodiversity is highest at the tropics
  - **Biodiversity hotspots**
    - Contain about 44% of known higher plant species and 35% of terrestrial vertebrate species
    - Represent only about 1.4% of earth's land area

Value of Biodiversity

- Direct Value
  - A great number of species perform services from which humans can derive an economic value.
  - These include, Medicinal value, Agricultural value, & Consumptive Use Value
  - **Medicinal Value**
    - Most of the prescription drugs currently used in the United States were originally derived from living organisms
      - Worth about $200 billion
      - Ex: Rosy Periwinkle – leukemia treatment; penicillin - treatment of bacterial infections; blood from Limulus - used to ensure that medical devices remain free of bacteria

Value of Biodiversity

- Direct Value
  - **Agricultural Value**
    - Wheat, corn, and rice are derived from wild plants that were modified to increase their yield
    - Natural predators of plant pests have been introduced to agricultural systems to reduce the impact of the pest on plant yields
  - **Consumptive Use Value**
    - Humans have had success cultivating crops, domesticating animals, growing trees in plantations, etc.
      - However, most freshwater and marine harvests must be hunted, rather than grown via aquaculture, for human consumption
    - Additional products associated with the environment are sold commercially
      - Wild fruits, vegetables, skins, fibers, beeswax and seaweed
      - Profits from the sale of these products provide an economic benefit to the human population

Value of Biodiversity

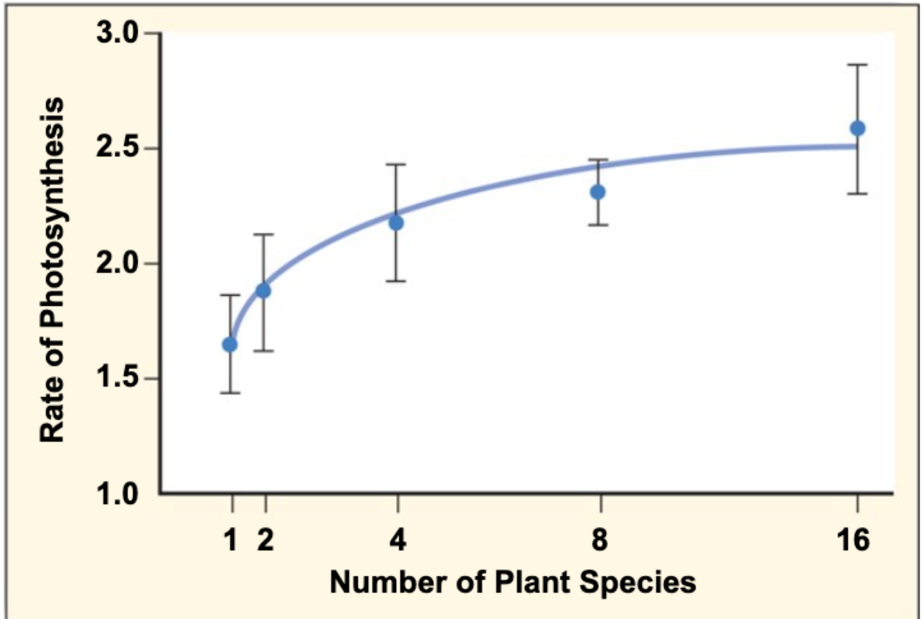
- **Indirect Value**
  - Based on the services ecosystems provide simply by their own existence.
  - These include Biogeochemical cycles, Waste recycling, Provision of Fresh Water, Prevention of Soil Erosion, Regulation of Climate, & Ecotourism
  - **Biogeochemical Cycles**
    - The biodiversity within ecosystems contributes to the workings of the water, carbon, nitrogen, phosphorous, and other biogeochemical cycles
    - Humans are dependent upon these cycles for fresh water, removal of carbon dioxide from the atmosphere, uptake of excess soil nitrogen, and provision of phosphate
  - **Waste Recycling**
    - Decomposers break down dead organic matter and other types of wastes into inorganic nutrients used by producers within ecosystems.
    - This function aids humans
      - The human population dumps millions of tons of waste material into natural ecosystems each year
      - If it were not for decomposition this waste would soon cover the entire surface of the Earth

Value of Biodiversity

- **Provision of Fresh Water**
  - The water cycle provides fresh water to terrestrial ecosystems
  - Humans use this fresh water in innumerable ways
  - Freshwater ecosystems provide us with a large diversity of species we can use as a source of food
  - Forests and some other natural ecosystems soak up water and release it at a regular rate, thereby reducing flooding
- **Prevention of Soil Erosion**
  - Intact ecosystems naturally retain soil and prevent soil erosion
  - The importance of this attribute is particularly observed after deforestation
- **Regulation of Climate**
  - Trees provide shade and reduce the need for fans and air conditioners in the summer
  - Globally, forests restore the climate by incorporating carbon dioxide from the atmosphere
  - Reduction of forests reduces the carbon dioxide uptake and oxygen output through Photosynthesis
- **Ecotourism**
  - In the United States, nearly $4 billion is spent on fees, travel, lodging, and food within natural settings
  - Many underdeveloped countries in tropical regions take advantage of this by offering “ecotours” of the local biodiversity

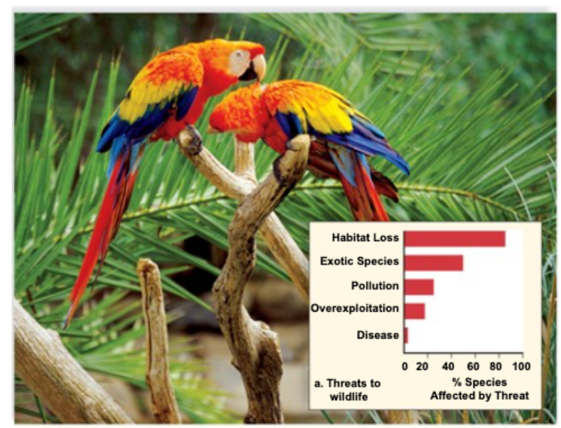
Value of Biodiversity

- Biodiversity and Natural Ecosystems
  - Large-scale changes in biodiversity have significant impacts on ecosystems:
    - Ecosystem performance improves with increasing biodiversity
    - Rate of photosynthesis increases as biodiversity increases

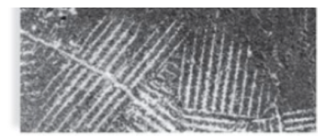


Causes of Extinction

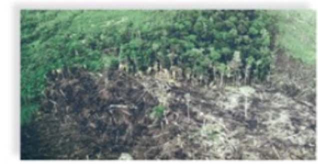
- Known causes of species extinction are due to:
  - Habitat loss (85%)
  - Exotic species (50%)
  - Pollution (24%)
  - Overexploitation (17%)
  - Disease (3%)
    - % of species affected by threat



b. Macaws


Roads cut through forest


Forest occurs in patches


Destroyed areas

- **Habitat Loss**
  - Occurs in all ecosystems
  - Recent concern focuses on tropical rain forests and coral reefs because they are rich in species
  - Loss of habitat affects terrestrial, freshwater, and marine biodiversity

Causes of Extinction

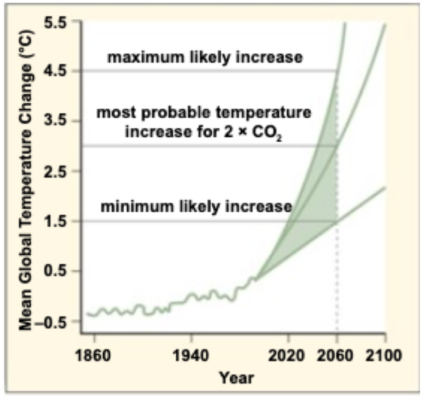
- **Exotic Species**
  - Nonnative species that migrate, or are introduced, into a new ecosystem
  - Humans introduce exotic species into ecosystems through:
    - Colonization, Horticulture and Agriculture, & Accidental Transport
  - Islands particularly susceptible, because of limited migration
    - Japanese Kudzu vines brought in to Southern US to stop erosion; overtake native plants
    - Mongoose introduced in Hawaii to control rat pop. Now prey on native birds

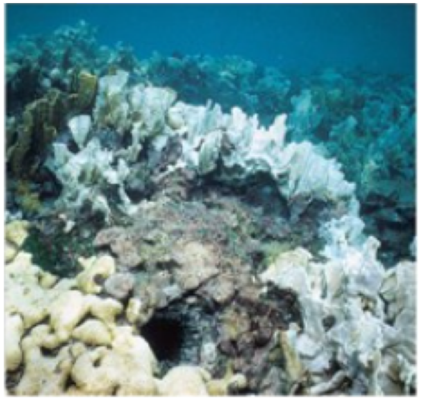
Causes of Extinction

- **Pollution**
  - Any environmental change that adversely affects living things
  - Third main cause of extinction
  - Biodiversity is particularly threatened by
    - Acid deposition
      - Sulfur and nitrogen depositions resulting from automobiles
        - Weakens trees and kills decomposers and small invertebrates
    - Eutrophication
      - Nutrient runoff causes algae to die, reduces oxygen of waterways
    - Ozone depletion
      - Chlorofluorocarbons break down protective layer
    - Synthetic organic chemicals
      - Products can mimic effect of hormones, and effect wildlife

Causes of Extinction

- **Climate Change**
  - Refers to recent changes in the Earth's climate
  - Increased temperature of the Earth results in drastic climatic changes
    - Temperature increase is caused, in part, by increased concentrations of greenhouse gases, such as CO_2 that serve to trap heat within the atmosphere
      - Results in ecosystem disruption and extinction





Causes of Extinction

- **Overexploitation**
  - The number of individuals taken from the population is so great that the population becomes severely reduced in numbers
  - Positive feedback cycle
    - The smaller the population, the more valuable its members, and the greater the incentive to capture the few remaining organisms
  - The market forces driving overexploitation:
    - Exotic Pets, Poaching, & Overfishing

a. Fishing by use of a drag net

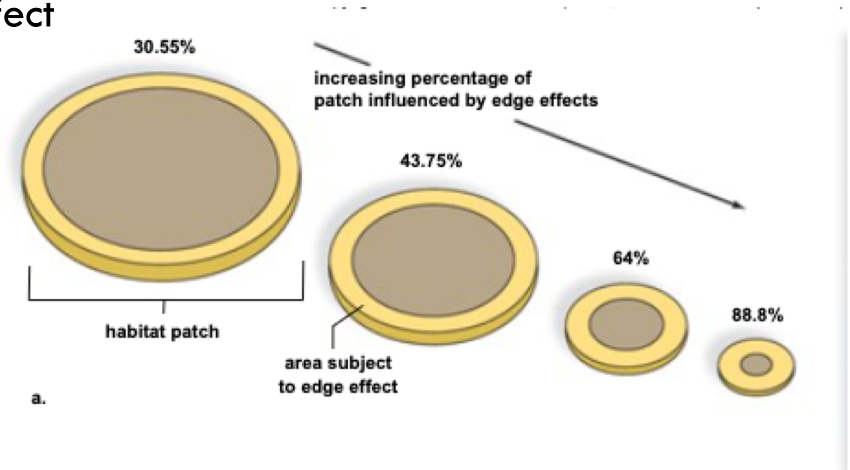
b. Result of drag net fishing

Conservation Techniques

- Habitat preservation and restoration are important in preserving biodiversity
- **Habitat Preservation**
  - Biodiversity hotspots, small areas with large numbers of endemic species not found anyplace else, are important targets for conservation
  - **Keystone Species**
    - Species that influence the viability of a community
    - Extinction of these species can lead to additional extinctions and loss of biodiversity
      - Ex -
  - **Flagstone Species**
    - Charismatic species that evoke a strong emotional response in humans
      - Ex -

Conservation Techniques

- **Landscape Preservation**
  - Landscape protection for one species benefits other wildlife in the same space
- **The Edge Effect**
  - The edge around a patch of habitat has conditions different from the patch interior
    - Forest edge vs interior of forest
  - An edge reduces the amount of habitat typical for an ecosystem
    - Can result in a significant reduction in population size
    - Creating a smaller patch increases the proportion of individuals subject to edge effect



Conservation Techniques

- Habitat Restoration
  - Restoration ecology seeks scientific ways to return ecosystems to their state prior
    - Three Principles of restoration ecology:
      - Begin as soon as possible before remaining fragments are lost
      - Once natural history is understood, use biological techniques to mimic natural processes
      - Goal is sustainable development